
Mechanics

Bionics Eng.

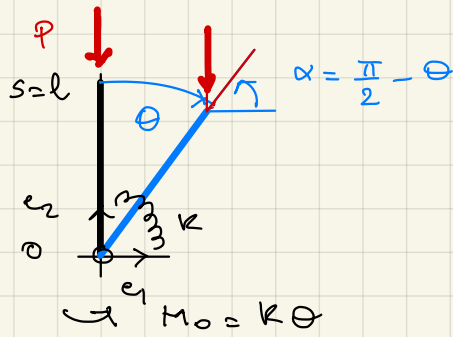
Lecture 19

2/12/2022



Lecture 19 Exercises on principle of minimal potential energy

30/x1/2020



Rigid rod, $E \rightarrow \infty$, $k = 0$

Elastic pin-joint at end $s = 0$

Dead load (= const. force) $\underline{F} = -P \underline{e}_2$, $P > 0$ at $s = l$

Find: equilibrium configurations and discuss their stability

We will confine analysis to $\Theta \in [-\pi, \pi]$

Some preliminary considerations.

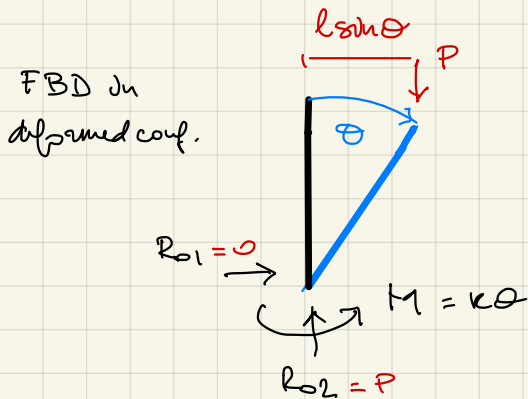
Kinematics

$\underline{\epsilon} = \underline{\kappa} = 0 \Rightarrow \alpha(s) \equiv \alpha(0)$ } only one scalar parameter describes config. (1 DoF system)
 $\underline{x}(0) = \underline{0}$

Change variable to $\Theta = \frac{\pi}{2} - \alpha \in [-\pi, \pi]$

Equil. config

To determine the equilibrium value Θ_{eq} for the angle Θ we need ad-hoc postulate



$$M_0 = k\Theta - Pl \sin\Theta = 0 \quad (*)$$

Solve for Θ to get Θ_{eq} , see later.

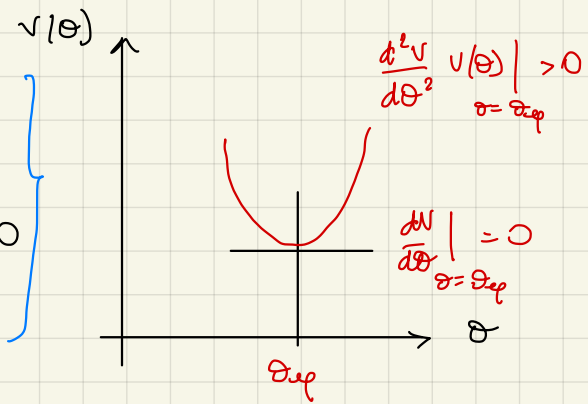
We'll arrive at the same equation (*) with an algorithm, based on the principle of minimal / stationary potential energy.

Define total potential energy $V(\theta) = V_{el}(\theta) - W_{ext}(\theta)$

Equilibria are stationary points of V : $\left. \frac{d}{d\theta} V(\theta) \right|_{\theta=\theta_{eq}} = 0$

If θ_{eq} is a local min for V , then it's stable: $\left. \frac{d^2}{d\theta^2} V(\theta) \right|_{\theta=\theta_{eq}} > 0$

[If θ_{eq} is a local max, then it's unstable]



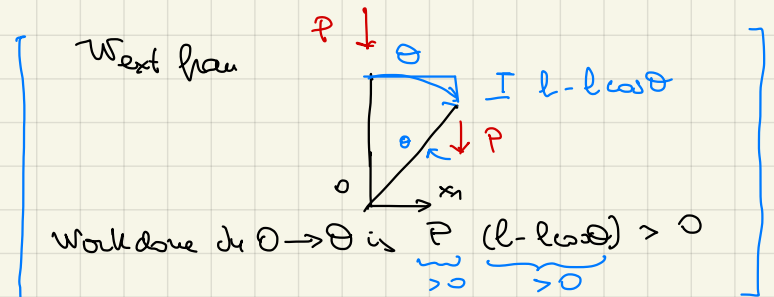
$$V(\theta) = \frac{1}{2} k\theta^2 - P(l - l\cos\theta)$$

$$+ Pl(\cos\theta - 1) \quad \rightarrow \text{independent constant}$$

$$\frac{dV}{d\theta} = k\theta - Pl\sin\theta$$

$$\theta_{eq} \text{ by solving } \left. \frac{dV}{d\theta}(\theta) \right|_{\theta=\theta_{eq}} = 0, \text{ i.e.}$$

$$k\theta - Pl\sin\theta = 0 \quad \text{as } (\rightarrow) \text{ seen before}$$



Alternatively, ΔW_{ext}

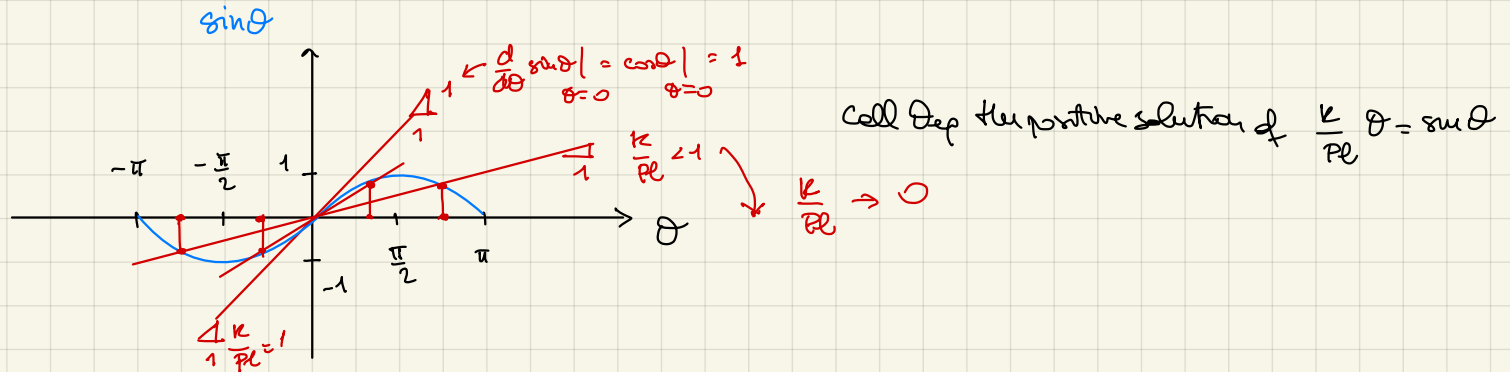
$$\begin{aligned} W_{ext} &= (-P\mathbf{e}_2) \cdot (\mathbf{x}_\theta(l\mathbf{e}_2) - \mathbf{x}^R(l\mathbf{e}_2)) \\ &= (0, -P) \cdot ((l\sin\theta, l\cos\theta) - (0, l)) \\ &= Pl - Pl\cos\theta \end{aligned}$$

To check for stability we'll need to check $\left. \frac{d^2}{d\theta^2} V(\theta) \right|_{\theta=\theta_{eq}} = (k - Pl\cos\theta) \Big|_{\theta=\theta_{eq}} \stackrel{\text{stable}}{\geq 0}$ unstable

Solve $\frac{d}{d\theta} V(\theta) = k\theta - Pl \sin\theta = 0$

$\theta = \frac{Pl}{k} \sin\theta \rightarrow \begin{cases} \sin\theta = 0 \\ \sin\theta \neq 0 \end{cases} \begin{cases} \theta = 0, \text{ equil } \forall P \\ \frac{k}{Pl} \theta = \sin\theta \neq 0, \exists \text{ solu's } \theta \neq 0 \text{ for } P = P_{cr}(\theta) = \frac{k}{l} \frac{\theta}{\sin\theta} \neq 0 \end{cases}$

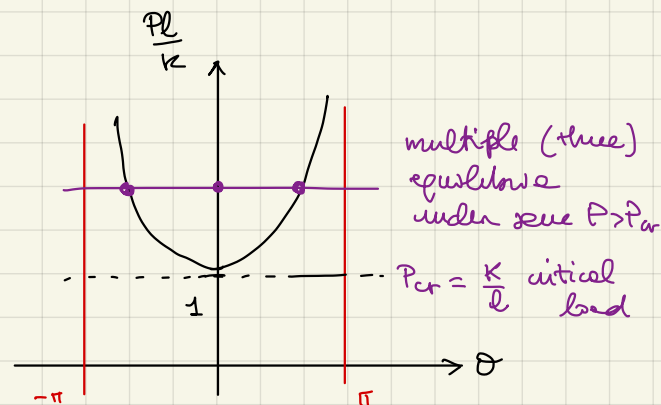
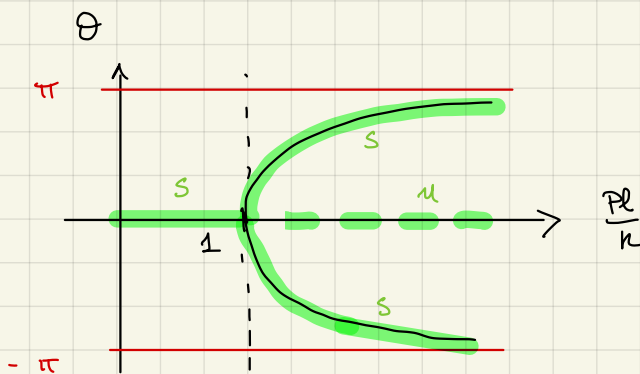
equilibrium values of θ can be obtained by solving $\frac{k}{Pl} \theta = \sin\theta$



For $\frac{Pl}{k} > 1$ there are two solutions of opposite sign $\pm \theta_{eq}$, tending to $\pm \pi$ as $\frac{Pl}{k} \rightarrow +\infty$

θ_{eq} as a function of P

P_{cr} as a function of θ_{eq}



Bifurcation diagrams

$P_{cr} = \frac{k}{l}$ critical load for lateral instability (bifurcation)

For $P > P_{cr}$ there are multiple (three) equilibria under the same load
which of these are stable?

